

ADFORGE AI — PROJECT DOCUMENTATION

Solution Requirements (Functional & Non-Functional)

- **Date:** 7th July 2026
- **Development Profile:** Sunehri Sharma
- **Maximum Marks:** 4 Marks

Step 1: Proposed Functional Requirements (FR)

S.No	Requirement	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Single-user local environment verification. The system uses secure environment validation to confirm local API configuration keys are loaded prior to rendering the interface workspace.	High
2	Authorization levels	Local Marketer Role: Grants full access to configure ad variables, execute generations, use quick-copy triggers, and perform real-time content refinements.	High
3	External interfaces	Seamless HTTPS connection to Groq's cloud API endpoint using the high-performance LLaMA 3.3-70B-Versatile model.	High
4	Transactions processing	Processes raw user input forms (USPs, products, platform checkboxes) into structured system prompt payloads, parses LLM stream outputs, and returns valid JSON datasets asynchronously.	High
5	Reporting	Interactive frontend output canvas displaying multi-platform ad variants, tag pills, emojis, slogans, and A/B test suggestions, with one-click copy and	Medium

S.No	Requirement	Requirement Description	Priority (High/Medium/Low)
		markdown export triggers.	
6	Business rules, etc.	Implements strict platform formatting constraints: Google Ads headlines must strictly remain within a 30-character limit; Instagram captions must render with structured emojis and categorized hashtags.	High
7	Compliance to laws	Secure, local API credential handling via standard .env configuration files to prevent API key leaks, adhering to strict data privacy principles by never writing user inputs to persistent databases.	Medium
8	Other (Refinement Loop)	Provides an active "Refine Copy" processing engine that updates selected outputs dynamically based on custom iterative instructions.	High

Step 2: Proposed Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Optimizes backend processing calls and LLM inference.	Ad copy configurations for all selected platforms must process and render within 3 seconds using Groq's high-speed API.
2	Scalability	The application's local architecture must support rapid traffic handling when exposed to external testing.	Supports seamless simultaneous active generation threads over local and secure pyngrok public tunnels.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	Ensures API endpoints and configurations are strictly locked.	Zero data persistence. No user prompts or API keys are written to disk. Key credentials are held purely in system environment memory.
4	Reliability & Availability	Fault-tolerant design for handling external network drops or API connectivity issues.	The Flask backend remains active on localhost:5000 with 99.9% local uptime, handling service disconnects gracefully using automated error-catch UI banners.
5	Usability & Accessibility	High-end visual interface designed to keep layouts and copy readable.	Features a modern dark-themed glassmorphism workspace UI that adjusts dynamically to desktop, tablet, and mobile browsers.
6	Maintainability & Portability	Streamlined local environment setup for developers and evaluators.	Complete, isolated app setup on any device within 2 minutes using standard requirements.txt dependencies.